CIS410 – Operating Systems

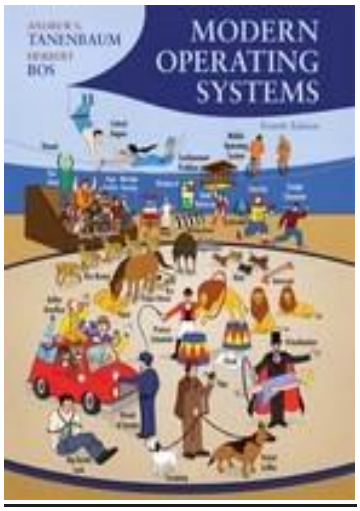
Course Syllabus

Course Description:

This course will provide depth and profound understanding of fundamental concepts of operating system and its use. Topics will cover operating system concepts including Operations, System Structure, Process Concept, Multithreaded Programming, Process Scheduling, Synchronization, Deadlocks, Memory-Management Strategies, Implementing File Systems, Secondary-Storage Structure, I/0 Systems, System Protection, System Security, Distributed Operating Systems, and Distributed File Systems.
Prerequisites: CIS380

Text Book

eTextbook:
Modern Operating Systems, Fourth Edition | 9780133591620
Author(s): Andrew S. Tanenbaum; Herbert Bos
Additional product details
eText:
ISBN-10 0133592472, ISBN-13 9780133592474
Print:
ISBN-10: 013359162X, ISBN-13: 9780133591620
Author(s): Andrew S. Tanenbaum; Herbert Bos
Publisher: Pearson
Copyright year: © 2015 Pages: 1136



Software Required:
Microsoft Visual Basic 2010 or 2012: (Microsoft Visual Studio 2010 or 2012, incorporates C++ 2010 or 2012): Download Microsoft Visual Studio it at no charge from Microsoft DreamSpark: https://www.dreamspark.com/

Learning Outcomes

1. Students will demonstrate knowledge in operations, process management, memory management, storage management, protection and security, and distributed systems.
2. Students will analyze various process management concepts, such as process scheduling, process synchronization, CPU scheduling, deadlock handling, scheduling algorithms, and threads.

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3. Students will define and describe how execution of a process manages main memory. The computer needs to retain several processes in the memory to improve both the utilization of the CPU and the speed of its response to its users.
4. Students will define and describe the storage management structures and how I/O, file and mass storage are handled in a computer system.
5. Students will identify and describe how unauthorized activities must be protected from another user and how operating system processes authorization to proper user to gain access to files, memory, CPU and other system resources.
6. Students will define and describe how distributive system deals with various deadlock problems and provides various process synchronization and communication mechanisms.
7. Also, define and describe how it allows computation speed to improve the data reliability and data availability by sharing resources.

Course Requirements

- View Presentations or Listen to Lectures
- Complete Assigned Readings
- Complete Programming Assignments
- Participate in Discussion Boards
- Complete Test/Quizzes/Projects

Grading Policy

Discussion Boards.....20%
Assignments…… ….....45%
Final Exam… …….....……..…35%
Total **100%**

Discussion Board Questions

The Discussion Board is an interactive tool that could be considered a “virtual classroom”. Its purpose is to allow students and instructors to interact. Each week a question will be posted to the Discussion Board. Answer the discussion question. Also reply to at least two of your colleague’s postings. (total of three postings for each thread: one initial and two responses). Your initial posting should be made no later than Wednesday of each week. Your final posting is due Sunday at midnight.

Assignment Submission – Programming Assignments

Post

In each learning unit, you may be required to submit assignments in the form of attachments. All assignments must be complete and returned in the Assignment Submission section of each Learning Unit. **All assignments are due by day 7 of the Unit's respective week, except for Unit 8**. This does not include the first posting to the discussion board, as noted below in the *Course Schedule*.

Learning Unit Final Exam

The Final Exam/Project will be administered during Week 8. Instructions are posted in Unit 8.

How to approach this class

Allow Time. Most students are not prepared for the amount of time that an on-line course takes. Therefore, each chapter must be read carefully. This takes time – more time than sitting in a classroom listening to a lecture.

Keep Up – Be Organized. Operating Systems is a course that builds as it goes. Concepts learned in one chapter are combined with concepts in subsequent chapters. Remember this is an accelerated degree program course. This course covers 16 weeks of material in 8 weeks.

Meet Deadlines. Do not get behind. Have each week's assignments completed on time.

Keep in Touch. During this course, it is very important that you keep in touch with me and with and your fellow classmates Your instructor and your peers cannot help you and you cannot help them if you are not around through e-mail. It's very important to keep the lines of communications open. So Keep in Touch!

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Plagiarism

The entire University community must adhere to the following policy. Academic dishonesty consists of plagiarism, cheating, unauthorized copying of computer discs, willful introduction of viruses, willful interruption of systems, and may also include unintentional complicity in others' academic dishonesty. Any reported cases will be directed to the Vice President for Academic Affairs and/or Dean of Faculty, who will refer these cases to an academic dishonesty panel, which will consist of faculty, staff, and

students. Confirmed dishonesty March result in the assignment of a failing grade or other penalties.

CIS410 Operating Systems
Course Schedule

Unit 1	
Course Activities / Textbook Chapters	
• View Presentations or Listen to Lectures • Complete Assigned Readings • Homework Assignments • Complete Programming Assignments • Participate in Discussion Boards	**Chapter 1: <u>Introduction</u>** **Assignment:** Read Chapter 1: Page(s) 1 - 81 **Problems:** Page(s) 81-82 Do Question(s) # 1, 3, 5, 10, 12, and 17 **Assignments are due by this Sunday, no later than midnight.**
Discussion Question	

Unit 2	
Quiz 1:	
• View Presentations or Listen to Lectures • Complete Assigned Readings • Homework Assignments • Complete Programming Assignments • Participate in Discussion Boards • Quiz 1	**Chapter 2: Process And Threads** **Chapter 3: Memory Management** **Assignment:** Read Chapter 2: Page(s) 85 - 173 **Problems:** Page(s) 174-178 Do Question(s) # 4, 14, 31, and 53 **Assignment:** Read Chapter 3: Page(s) 181- 254 **Problems:** Page(s) 254 – 260 Do Question(s) # 4, 13, and 48 **Quiz 1:** Chapter 1 and 3

Post

	Assignments are due by this Sunday, no later than midnight.
Discussion Question	

Unit 3	
Project:	
• View Presentations or Listen to Lectures • Complete Assigned Readings • Homework Assignments • Complete Programming Assignments • Participate in Discussion Boards • Project	**Chapter 4: File Systems** **Assignment:** Read Chapter 4: Page(s) 263 - 332) **Problems:** Page(s) 333 - 335 Do Question(s) # 4, 5, 6, 7, and 33 **<u>Project:</u>** **Assignments are due by this Sunday, no later than midnight.**
Discussion Question	

Unit 4	
Midterm Exam:	
• View Presentations or Listen to Lectures • Complete Assigned Readings • Homework Assignments • Complete Programming Assignments • Participate in Discussion Boards • Midterm Exam	**Chapter 5: Input/Output** **Assignment:** Read Chapter 5: Page(s) 337 - 429) **Problems:** Page(s) 429 - 431 Do Question(s) # 4, 5, 13, and 16 **<u>Midterm Exam:</u>** Chapter 1- 5 **Assignments are due by this Sunday, no later than midnight.**
Discussion Question	

Unit 5	
Course Activities / Textbook Chapters	
• View Presentations or Listen to Lectures • Complete Assigned Readings • Homework Assignments • Complete Programming Assignments	**Chapter 6: Deadlocks** **Assignment:** Read Chapter 6: Page(s) 435 - 465) **Problems:** Page(S) 465 - 467 and 515

Post

• Participate in Discussion Boards • Complete Test/Quizzes	Do Question(s) # 1, 8, 22, and 24 **Assignments are due by this Sunday, no later than midnight.**
Discussion Question	

Unit 6	
Course Activities / Textbook Chapters	
• View Presentations or Listen to Lectures • Complete Assigned Readings • Homework Assignments • Complete Programming Assignments • Participate in Discussion Boards • Complete Test/Quizzes • Quiz 2	**Chapter 7: Virtualization and the Cloud** **Chapter 8: Multiple Processor Systems** **Assignment:** Read Chapter 7: Page(s) 471 - 515) **Problems:** Page(s) 515 Do Question(s) # 1, 2, , 4, 8, and 9 **Assignment:** Read Chapter 8: Page(s) 517 – 589) **Problems:** Page(s) 589 - 591 Do Question(s) # 4, 24, and 27 **Assignments are due by this Sunday, no later than midnight.**
Discussion Question	

Unit 7	
Project Submission:	
• View Presentations or Listen to Lectures • Complete Assigned Readings • Homework Assignments • Complete Programming Assignments • Participate in Discussion Boards • Project submission	**Chapter 9: Security** **Chapter 12: Operating Systems Design** ***Assignment:*** *Read Chapter 9: Page(s) 593 - 776)* ***Problems:*** *Page(s) 706 - 710* *Do Question(s) # 6, 14, and 38* ***Read Chapter 12:*** *Page(s) 982 - 1027* ***Problems:*** *Page(s) 1027 - 1029* *Do Question(s) # 9, 19, and 25* **Assignments are due by this Sunday, no later than midnight.**
Discussion Question	

Unit 8
Final Exam:

Post

• View Presentations or Listen to Lectures • Complete Assigned Readings • Homework Assignments • Complete Programming Assignments • Participate in Discussion Boards • Complete Test/Quizzes • Final Exam	**Chapter 10: Case Study 1 Unix, Linux, and Android** **Assignment:** Read Chapter 10: Page(s) 713 - 850) **Problems:** Page(s) 851 - 854 Do Question(s) # 1, 7, 16, and 47 **<u>Final Exam</u>: 6 - 10, and 12** **Assignments are due Friday, and Final Exam is due Saturday, no later than midnight.**
Discussion Question	